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EXPERIMENT 3

NETWORK SECURITY ESSENTIALS

AIM:

To observe how basic network security works and how devices protect a network from attacks.

ALGORITHM:

1. Open the network security essentials.
2. Observe the network diagram and devices.
3. See how data moves between devices.
4. Notice how the firewall/IDS blocks malicious traffic.
5. Complete the given tasks.

Task 1 Introduction

The networks are the backbone of every modern organization. Servers, workstations, applications, and security devices don't exist in isolation; they are interconnected, forming a single ecosystem. The network perimeter is where this internal ecosystem is segregated from the external Internet, and it is often the first target for attackers.

This room is designed to introduce you to the fundamental concepts of network security from a defensive perspective. While our focus will be on the logs and data sources we encounter daily, we'll start by building a strong understanding of how networks are structured, why they are important to protect, and why monitoring the perimeter is critical for early attack detection.

Learning Objectives

In this room, we will cover the following learning objectives:

- Understand what a network is and identify its key components.
- Explore the concept of the network perimeter and its importance.
- Identify the key perimeter threats.
- Examine the firewall logs to monitor normal and suspicious logs.

Prerequisites

This room expects the users to have covered the following rooms and know the basic networking concepts.

- [Network Fundamentals](#)
- [Intro to SIEM](#)
- [Network Security](#)
- [Intro to Log Analysis](#)

Let's begin our learning journey.

Answer the questions below

Complete the task.

No answer needed

✓ Correct Answer



Task 2 Lab Connection



Before moving forward, start the lab by clicking the **Start Machine** button. It will take 3 minutes to load properly. The VM will be accessible on the right side of the split screen. If the VM is not visible, use the blue Show Split View button at the top of the page.

Set up your virtual environment

To successfully complete this room, you'll need to set up your virtual environment. This involves starting the Target Machine, ensuring you're equipped with the necessary tools and access to tackle the challenges ahead.



Target machine

Status:  Off

 Start Machine

All files required to complete the room are placed in the **Perimeter_logs** folder on the Desktop. The log files to be used in the task 7 for the investigation are pre-ingested into Splunk Instance, which can be accessed from within the Machine at **MACHINE_IP:8000**.

Answer the questions below

Connect to the Lab.

No answer needed

 Correct Answer

Task 3 A Network - Overview



Before understanding the security aspect of a Network, it is important to understand what a network is, what components or assets form it, and how they work.

Network Components - Building Blocks of a Network

A computer network is not just a random collection of devices; it is an organized structure where network assets connect to enable communication, resource sharing, and connectivity with each other and the world.

Small Enterprise Network

From a security perspective, knowing what these devices do and why they're important helps quickly identify suspicious activity. Let's take an example of a small enterprise network, and understand its key components, usage, and their importance from a security perspective:

Answer the questions below

Continue to the Next task.

No answer needed

 Correct Answer

Task 4 Network Visibility



Network visibility is essential in cyber security. It's the ability to monitor and understand what's happening across your network. It's a core principle for security analysts: you can't defend what you can't see. Effective visibility enables threat detection, incident investigation, and a strong security posture. It's not about tracking every packet but having the tools to build a clear picture. Without it, you're blind to potential threats.

The network visibility comes from two primary sources of logs. You must understand the difference between them to piece together an attack timeline effectively.

Answer the questions below

Continue to the next task.

No answer needed

 Correct Answer

Task 5 Network Perimeter

The network perimeter is the boundary that separates an organization's internal network (trusted zone, like employees, servers, business applications) from the external Internet (untrusted zone). It's the point where data enters or leaves the network. Think of it as the main gate or front door of a secure building. All traffic coming from the Internet must pass through this point to enter your network, and all internal traffic must pass through it to exit the Internet. The network perimeter is your first and most critical line of defense.

- The internal network is where business-critical systems live.
- The external Internet is full of potential threats.
- The perimeter is the controlled entry point through which all traffic passes.

Understanding the perimeter, its existence, and how it can be defended is critical for a security analyst.

Answer the questions below

Continue to the next task.

No answer needed

✓ Correct Answer

Task 6 Network Perimeters: Monitoring and Protecting

In the previous task, we discussed the network perimeter, its importance, and how it functions as the boundary between trusted internal systems and the external Internet. Attackers constantly probe this boundary, looking for weak spots. Let's explore various scenarios and how to monitor different network perimeters to identify threats and protect against them.

Monitoring the perimeter

Monitoring the perimeter means using firewalls, intrusion detection/prevention systems (IDS/IPS), and access control to examine and limit exposure and enforce security rules. This allows the security analysts to:

- Spot early-stage attacks like port scans or brute-force attempts.
- Detect misconfigurations that leave sensitive services exposed.
- Identify outbound traffic anomalies that may indicate malware or data exfiltration.

The perimeter logs used in the following scenarios can be found at `/Perimeter_logs/task6/` folder on the Desktop.

Answer the questions below

Examine the firewall logs. Which IP address is performing the port scan?

203.0.113.10

✓ Correct Answer

In the WAF Logs, which single source IP is responsible for all the blocked web attacks?

198.51.100.12

✓ Correct Answer

In the VPN logs, how many brute-force attempts failed?

90

✓ Correct Answer

Which suspicious IP address was found attempting the brute-force attack against the VPN gateway?

45.137.22.13

✓ Correct Answer

Task 7 Perimeter Logs: Investigating the Breach

Incident Scenario

Initech Corp, a mid-sized financial services company, has recently deployed a new firewall and intrusion detection system (IDS) to monitor its network perimeter. Over the past month, security analysts have noticed abnormal traffic patterns, but the SOC team has been overwhelmed and missed deeper analysis.

As a new security analyst, you have been tasked with reviewing one month of perimeter logs to determine what techniques the adversary used, and whether they succeeded in breaching the perimeter.

You have been given three sets of logs from the time of the incident. The logs can be found in the `Perimeter_logs/challenge` directory on the Desktop.

- Firewall Logs: `firewall.log`
- WAF Logs: `ids_alerts.log`
- VPN Logs: `vpn_auth.log`

Answer the questions below

Examine the firewall logs. What external IP performed the most reconnaissance?

203.0.113.45

✓ Correct Answer

In the firewall log, Which internal host was targeted by scans?

10.0.0.20

✓ Correct Answer

Which username was targeted in VPN logs?

svc_backup

✓ Correct Answer

What internal IP was assigned after successful VPN login?

10.8.0.23

✓ Correct Answer

Which port was used for lateral SMB attempts?

445

✓ Correct Answer

In the IDS logs, which host beaconed to the C2?

10.0.0.60

✓ Correct Answer

During the investigation, which IP was observed to be associated with C2?

198.51.100.77

✓ Correct Answer

Which host showed the exfiltration attempts?

10.0.0.51

✓ Correct Answer

Task 8 Conclusion

In this room, you learned that an enterprise network is not just a collection of computers and servers. It's an ecosystem with critical components like firewalls, Active Directory, application servers, file/database servers, endpoints, and wireless access points.

- The network perimeter acts as the boundary between trusted and untrusted networks.
- Attackers test this boundary constantly with port scans, brute-force attempts, and exploits against exposed services.
- Monitoring the perimeter allows SOC analysts to detect these early signs before attackers can move deeper into the network.

As a security analyst, our role is to:

- Recognize normal vs suspicious network traffic.
- Escalate unusual activity to higher SOC tiers.
- Understand how each network component fits into the bigger picture of enterprise defense.

Answer the questions below

Complete the room.

No answer needed

✓ Correct Answer

RESULT:

Understood how network security devices protect the network and detect attacks.